## Exercise - 1.5

Choose the correct or the most suitable answer.

1. If $A = \{(x, y) : y = e^x, x \in R\}$ and $B = \{(x, y) : y = e^{-x}, x \in R\}$ then $n(A \cap B)$ is
   (1) Infinity (2) 0 (3) 1 (4) 2

2. If $A = \{(x, y) : y = \sin x, x \in R\}$ and $B = \{(x, y) : y = \cos x, x \in R\}$ then $A \cap B$ contains
   (1) no element (2) infinitely many elements
   (3) only one element (4) cannot be determined.

3. The relation $R$ defined on a set $A = \{0, -1, 1, 2\}$ by $xRy$ if $|x^2 + y^2| \leq 2$, then which one of the following is true?
   (1) $R = \{(0, 0), (0, -1), (0, 1), (-1, 0), (-1, 1), (1, 2), (1, 0)\}$
   (2) $R^{-1} = \{(0, 0), (0, -1), (0, 1), (-1, 0), (1, 0)\}$
   (3) Domain of $R$ is $\{0, -1, 1, 2\}$
   (4) Range of $R$ is $\{0, -1, 1\}$

4. If $f(x) = |x - 2| + |x + 2|, x \in \mathbb{R}$, then

   (1) $f(x) = \begin{cases} -2x & \text{if} \quad x \in (-\infty, -2] \\ 4 & \text{if} \quad x \in (-2, 2] \\ 2x & \text{if} \quad x \in (2, \infty) \end{cases}$

   (2) $f(x) = \begin{cases} 2x & \text{if} \quad x \in (-\infty, -2] \\ 4x & \text{if} \quad x \in (-2, 2] \\ -2x & \text{if} \quad x \in (2, \infty) \end{cases}$

   (3) $f(x) = \begin{cases} -2x & \text{if} \quad x \in (-\infty, -2] \\ -4x & \text{if} \quad x \in (-2, 2] \\ 2x & \text{if} \quad x \in (2, \infty) \end{cases}$

   (4) $f(x) = \begin{cases} -2x & \text{if} \quad x \in (-\infty, -2] \\ 2x & \text{if} \quad x \in (-2, 2] \\ 2x & \text{if} \quad x \in (2, \infty) \end{cases}$

5. Let $\mathbb{R}$ be the set of all real numbers. Consider the following subsets of the plane $\mathbb{R} \times \mathbb{R}$:

   $S = \{(x, y) : y = x + 1 \text{ and } 0 < x < 2\}$ and $T = \{(x, y) : x - y \text{ is an integer }\}$

   Then which of the following is true?
   (1) $T$ is an equivalence relation but $S$ is not an equivalence relation.
   (2) Neither $S$ nor $T$ is an equivalence relation
   (3) Both $S$ and $T$ are equivalence relation
   (4) $S$ is an equivalence relation but $T$ is not an equivalence relation.

6. Let $A$ and $B$ be subsets of the universal set $\mathbb{N}$, the set of natural numbers. Then $A' \cup [(A \cap B) \cup B']$ is
   (1) $A$ (2) $A'$ (3) $B$ (4) $\mathbb{N}$

7. The number of students who take both the subjects Mathematics and Chemistry is 70. This represents 10% of the enrollment in Mathematics and 14% of the enrollment in Chemistry. The number of students take at least one of these two subjects, is
   (1) 1120 (2) 1130 (3) 1100 (4) insufficient data